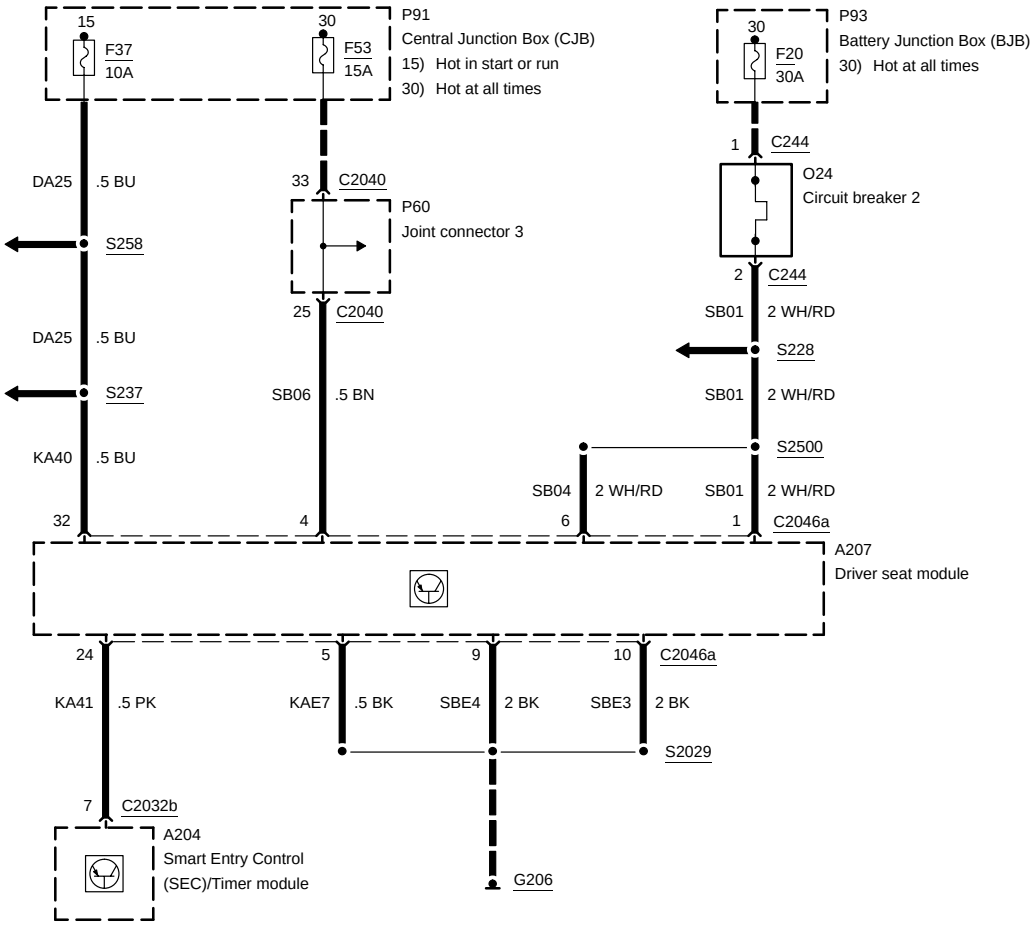with memory -2